

Unipolar Back-Projection Algorithm for Photoacoustic Tomography

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Abstract— Among different transducer configurations of a photoacoustic (PA) imaging system, limited-view configurations (e.g., planar and limited-view circular configurations) have more clinical applications. On the other hand, the generated PA wave is bipolar. Therefore, qualifying physiological parameters would be challenging, especially in a limited-view configuration. To tackle this issue, in this paper a unipolar back-projection reconstruction algorithm is presented. The proposed method uses the initial PA signals and derivatives to reconstruct a unipolar image. The numerical results showed that by using the proposed method, the quality of the reconstructed image by using only a half-view circular transducer array would be close to that of the full-view circular array. Quantitatively speaking, the generalized contrast-to-noise ratio (gCNR) value in the half-view configuration reaches 0.96. Moreover, the proposed method improves the resolution/contrast and gCNR values by ~ 50 %/15 dB and ~ 50%, respectively, compared to the conventional method.

Keywords—Photoacoustic, Back-projection, Optical absorption, Bipolar waves, Tomography.

I. INTRODUCTION

In a photoacoustic (PA) imaging (PAI) system, first, a laser source irradiates the biological tissue. Based on the optical absorption coefficient of the tissue, the tissue absorbs the light energy and thermal expansion occurs [1, 2]. A PA wave is then generated in all directions and would be captured by acoustic detectors [3]. The more solid angles of this generated PA wave are captured by the detectors, the higher quality map in terms of resolution and contrast of the optical absorption coefficient can be reconstructed [4, 5]. Therefore, among different PAI system transducer configurations, the full-view circular configuration has better resolution and contrast than the others [6-9].

The optical absorption coefficient is non-negative, but the generated PA wave is bipolar (positive followed by negative) [10]. Therefore, the reconstructed image contains positive and negative values (N-shaped). Qualifying physiological parameters (e.g., blood oxygen saturation (sO_2) and the metabolic rate of oxygen (MRO_2)) would be challenging in a bipolar image [11]. If the PAI system transducer configuration is full-view circular, the positive and negative parts of each reconstructed PA absorber will be symmetrical. Therefore, the bipolarity of the reconstructed image would be a minor issue. In planar and limited-view configurations, due to the capturing the solid angles of the generated PA waves in a few angles, the streak artifacts appear [12, 13], and also the positive and negative pixels in the reconstructed map of

the excited PA absorber will not be symmetric. This asymmetry makes the medical interpretation and qualification of physiological parameters of the biological image difficult.

To tackle the bipolar issue and to achieve a higher quality reconstructed image, some methods have been reported. In planar configuration, first, the Hilbert transformation is taken along the axial direction. Since the pixels have a complex value, the absolute value of each pixel is calculated [14-16]. Although applying Hilbert transform extracts the envelope of the reconstructed A-line, due to the limited bandwidth of the transducer, it will extend the point spread function (PSF). A simple method is to keep only the positive values and zero other values. But this method removes useful structures in the reconstructed image compared to the ground truth and induces artifacts [17-19]. A signal restoration algorithm has been proposed to compensate for the bipolar issue and improve the reconstructed image quality [20]. Iteration-based algorithms with a nonnegativity constraint have also been developed to tackle the bipolar issue [21, 22]. Still, these algorithms are time-consuming and require accurate modeling of the imaging system. This especially is challenging in heterogeneous medium. Li, et al. [11] developed a multi-view Hilbert transform (MVHT) method to solve the bipolarity issue. Although a unipolar reconstructed image is obtained by MVTH, the resolution of the MVTH method is slightly worse than the bipolar reconstructed image.

In this paper, to solve the bipolarity in planar and limited-view circular transducer configurations, a unipolar back-projection method is presented. In the proposed method both the initial and derivative of the captured PA signal are required for the back-projection. The results showed that in addition to the unipolar reconstructed image, the proposed method improves the resolution and edge uniformity compared to the conventional method. Section II is the detail of the proposed method. The results and discussion are in section III, and the conclusion is in section IV.

II. MATERIALS AND METHODS

As stated before, the generated wave from each excited PA absorber would be N-shaped (see Fig. 1(a1, b1)). Therefore, in planar and limited-view circular configurations, the reconstructed image by the delay-and-sum (DAS) method [23, 24] contains positive and negative values. In the captured signal by the transducer, the slope of the signal between the

positive and following negative peaks is a negative value. Therefore, in back-projection, if the derivative of each captured PA signal by a transducer is multiplied by -1, a new PA signal is generated (see Fig. 1(a3)). We expect that, the reconstructed image of the mentioned new signals with some modifications can compensate the negative parts in conventional reconstructed image. The detailed procedure of the proposed method is explained in the following steps:

1. Reconstructing an initial bipolar image with initial PA signals by performing the DAS algorithm (see Fig. 1(a2, b2)).
2. Generating new signals by multiplying the derivative of each initial signal by -1.
3. In the generated signals, the slope of the parts that refer to the location of the excited PA absorbers are almost constant. To compensate the negative slope of the initial bipolar image, a positive dynamic time-based slope (t) is multiplied to the generated signals.
4. Reconstructing a secondary bipolar image with generated PA signals (see Fig. 1(a3, b3)).
5. Normalizing both the initial and secondary reconstructed images and then summing them together as a new reconstructed image.
6. Uni-polarizing of the new reconstructed image by keeping the positive values of the new reconstructed image and zeroing negative values (see Fig. 1(a5, b5)).

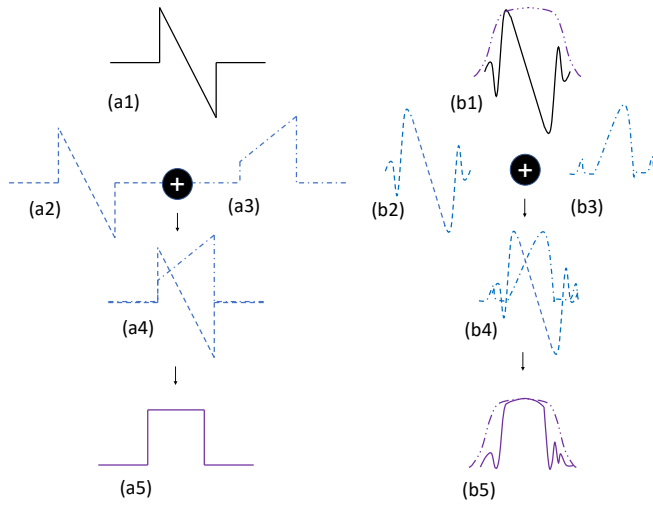


Fig. 1. (a1) a captured PA signal by an infinite bandwidth detector. (b1) a captured PA signal by a band-limited detector. The solid-dotted line indicate the envelope. (a2, b2) the back-projected initial signals. (a3, b3) the back-projected new generated signals. (a4, b4) summation of two back-projected signals. (a5, b5) extracting positive values of (a4, b4).

III. NUMERICAL STUDY AND RESULTS

A. Numerical Study

The numerical PA signals were synthesized by the k -Wave MATLAB toolbox [25]. Two types of limited-view transducer configurations were considered as numerical studies; 1) a planar transducer configuration and 2) a half-view circular transducer configuration. Grid size was considered 0.025 mm in both X-Y directions in all studies. The number of acoustic transducers was considered 180 and 128 for half-view circular and planar configurations, respectively. The central frequency

and bandwidth of synthesized PA signals were 6.25 MHz and 66%, respectively, and the speed of sound was considered 1500 m/s. In planar and half-view circular configurations, point targets and Derenzo phantom were used as numerical phantom studies, respectively. For quantitative evaluation, axial resolution and generalized contrast-to-noise ratio (gCNR) [26] were measured in the reconstructed images by conventional and proposed methods.

B. Numerical Results

1) Case 1: Planar configuration

The numerical results of universal back-projection (UBP) with Hilbert transformation (HT) and the proposed methods are shown in Fig. 3 in a dynamic range of 60 dB. The red solid lines in Fig. 2(a1, b1) indicate the position of the transducers. The areas within the green solid boxes are magnified in Fig. (a2, b2). The proposed method improves the resolution and contrast compared to the conventional method (compare the resolution in Fig. 2(a2, b2)). Quantitatively speaking, the resolution for a 0.2 mm diameter PA absorber, reaches to 0.22, and 0.5 for the proposed and conventional methods, respectively. Moreover, the contrast ratio (CR) reaches to 50 and 35 dB in the proposed and conventional methods, respectively.

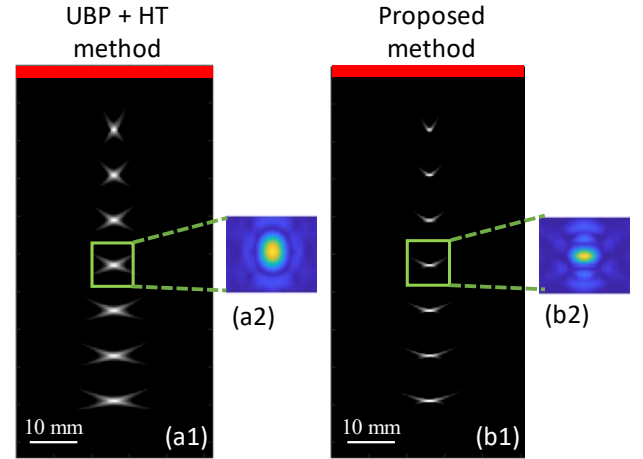


Fig. 2. The numerical results of point targets in a planar transducer configuration. The red solid line indicate the location of the transducer array. (a1) the reconstructed image by UBP method and applying Hilbert transformation. (a2) the magnified area within the green box in (a1). (b1) the reconstructed image by the proposed method. (b2) the magnified area within the green box in (b1).

2) Case 2: Half-view circular configuration

For half-view circular configuration, the results of three different reconstruction algorithms are compared with that of the proposed method. The numerical results of UBP, MVHT, keeping only the positive values and zeroing other values method (positive method) and the proposed method are shown in Fig. 3. In Fig. 3, where red half circles indicate the location of transducers. The results show that UBP reconstructs the image with low contrast. Furthermore, some major structures are missed in the positive method. Although the missing data is not occurring in the MVHT method, MVHT blurs the reconstructed image. The proposed method, reconstructs the map of optical absorption coefficient with high contrast and without any missing data (compare white arrows in Fig. 3). Quantitatively speaking, the gCNR values

for the UBP, MVHT, positive, and the proposed method reaches 0.64, 0.77, 0.30 and 0.96, respectively.

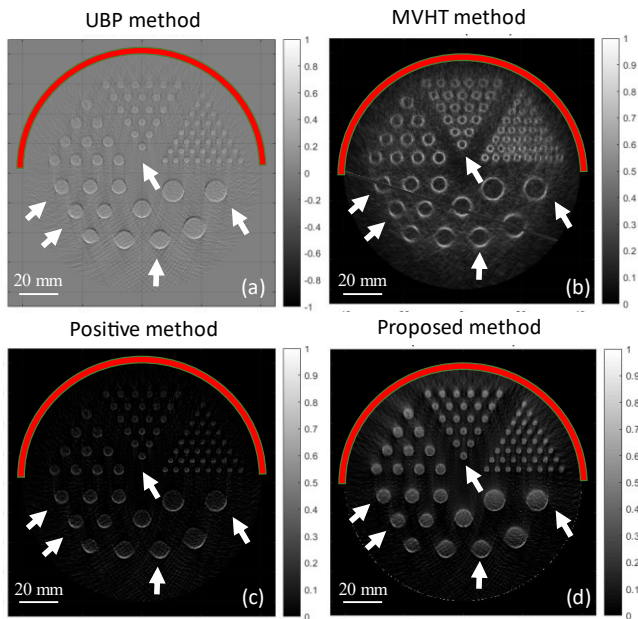


Fig. 3. The numerical results of Derenzo phantom in a half-view circular transducer configuration. The red solid curve indicate the location of the transducer array. (a, b, c, d) are the reconstructed image by UBP, MVHT, positive and the proposed method, respectively.

IV. CONCLUSION

In this paper, a unipolar back-projection reconstruction algorithm for limited-view transducer configurations in a PAI system is presented. In the proposed method, based on the derivative of captured PA signals by transducers, new PA signals are generated. Then, two images are reconstructed based on the captured PA signals and newly generated PA signals. By adding these two reconstructed images, and zeroing negative pixels, a unipolar reconstructed image is obtained. A planar transducer configuration and a half-view circular transducer configuration were used for numerical studies. The results show that the proposed method in comparison to the UBP method improves the resolution and contrast by $\sim 50\%$ and 15 dB, respectively. Also, the gCNR value in the proposed method reaches 0.96 which is $\sim 50\%$ more than the UBP method. The results show that by using only a half-view circular array of transducers and performing the proposed method, the non-negative reconstructed image will have a quality close to the same image obtained in the full-view circular array configuration. Moreover, in order to suppress artifacts, in future study a coprime sparse array method [27] can be applied to the reconstruction algorithms.

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